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Statistical Inference

The science of statistics deals with drawing conclusions from observed data.
We start with a large collection of data called a population, that have a measurable value associated with them.
By suitably sampling from this collection and then analyzing the sampled items, we hope to draw some conclusions about the distribution of the entire collection as a whole.

Markov's Inequality

If X is a random variable that only takes non-negative values, then for any a>0,

$$P\{X > a\} \leq \frac{E(X)}{a}$$

Proof:

For $a>0$, let an indicator be defined as $I = \begin{cases} 1, & x \geq a \\ 0, & \text{otherwise} \end{cases}$

since $X \geq 0$, $\quad I \leq \frac{x}{a}$

$$\Rightarrow E[I] \leq E\left[\frac{X}{a}\right]$$

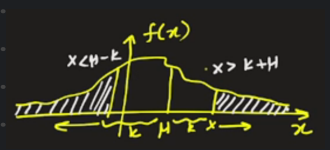
$$\Rightarrow E[I] \leq \frac{E[X]}{a}$$

$$\Rightarrow P\{X \geq a\} \leq \frac{E(X)}{a}$$

Chebyshev's Inequality

If X is a random variable with a finite mean μ and variance σ^2, then for any value k>0,

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$



Basically tells us that the value of X doesn't go too far from the mean.

Proof: since X is a random variable, $(X-\mu)^2$ is a non-negative r.v.
markov's inequality,

$$P\{(X-\mu)^2 \geq k^2\} \leq \frac{E[(X-\mu)^2]}{k^2}$$

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

These inequalities enable us to derive bounds on probabilities when only mean, or both the mean and variance are known.
If the actual distribution is known, the desired distributions can be calculated exactly and we won't need these bounds.

Q. If X is a continuous uniformly distributed random variable in (0, 10), what is the probability of |X-5| > 4?

→

$X \sim U_{[0,10]}$

using chebyshev's inequality,

$$P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

$$P\{|X-5| \geq 4\} \leq \frac{25}{3 \cdot 16}$$

$$\leq 0.52$$

$$\sigma^2 = \frac{(10-0)^2}{12} = \frac{100}{12} = \frac{25}{3}$$



Here, $|X-5| \geq 4 \Rightarrow X > 9$ or $X < 1$

$$P\{|X-5| \geq 4\} = P\{9 < X < 10\} + P\{0 < X < 1\}$$

$$= 0.1 + 0.1$$

$$= 0.2$$

Q. If X is a continuous normally distributed random variable (μ, σ^2), what is the probability of |X-μ| > 2σ? (given φ(2) = 0.9772)

→ Using Chebyshev inequality, $P\{|x-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}$

$$P\{|x-\mu| \geq 2\sigma\} \leq \frac{\sigma^2}{4\sigma^2}$$

$$\leq 0.25$$

$X \sim N(\mu, \sigma^2)$

$$P\{|x-\mu| \geq 2\sigma\} = P\left\{\frac{|x-\mu|}{\sigma} \geq 2\right\}$$

$$= P\left\{\frac{x-\mu}{\sigma} \geq 2 \text{ or } \frac{x-\mu}{\sigma} \leq -2\right\}$$

$$= P\{z > 2\} + P\{z < -2\}$$

$$= 1 - \phi(2) + \phi(-2)$$

$$= 1 - \phi(2) + 1 - \phi(2)$$

$$= 2(1 - 0.9772)$$

$$= 0.0456$$

So the bound isn't that good, but if the distribution isn't known, then the bound becomes useful.

Q. Suppose that it is known that the # of items produced in a factory during a week is a rv with mean 50.

a. What can be said about the probability that this week's production will exceed 75?
b. If the variance of this week's production is 25, then what can be said about the probability that this week's production will be between 30 and 70?

→ $\mu = 50$

a. Using Markov's inequality, $P\{X \geq a\} \leq \frac{E[X]}{a}$

$$\Rightarrow P\{X \geq a\} \leq \frac{50}{75}$$

$$\leq \frac{2}{3}$$

b. $P\{30 < X < 70\} = P\{|X-50| < 20\}$

Using Chebyshev's inequality,

$$P\{|X-50| \geq 20\} \leq \frac{\sigma^2}{k^2} = \frac{25}{20^2} = \frac{1}{16}$$

$$\Rightarrow -P\{|X-50| \geq 20\} \geq -\frac{1}{16}$$

$$\Rightarrow 1 - P\{|X-50| \geq 20\} \geq 1 - \frac{1}{16}$$

$$\Rightarrow P\{|X-50| < 20\} \geq \frac{15}{16}$$

Proposition:
If var(x) = 0, then P{X=E[X]} = 1.

The only random variables having variances equal to 0 are those which are constants with probability 1 at value of mean.

Proof: To prove: $P\{|X-\mu| > 0\} = 0$

$$P\{|X-\mu| > k\} \leq \frac{\sigma^2}{k^2}$$

$$P\left\{|X-\mu| > \frac{1}{n}\right\} \leq n^2\sigma^2$$

as $n \to \infty$, $\frac{1}{n} \to 0$

$$P\left\{\lim_{n\to\infty} |X-\mu| > \frac{1}{n}\right\} = 0$$

// Prob. values have to be continuous betn 0 & 1. $0 \times \infty \to 0$

$$P\{X \neq \mu\} = 0 \Rightarrow P\{X-\mu\} = 1$$

Sample mean:
Consider a population with a numerical value attached to it.
The population has a mean μ and a variance σ^2 (called the population mean and population variance).

Let x1, x2, ..., xn be a sample of values drawn from the population. $E[X_i] = \mu$, $E[(x_i-\mu)^2] = \sigma^2$.

$$\left\{\begin{matrix}\text{sample}\\ \text{mean}\end{matrix}\right\} : \quad \bar{x} = \frac{x_1 + \cdots + x_n}{n}$$

$$E[\bar{x}] = E\left[\frac{x_1+\cdots+x_n}{n}\right] = \frac{1}{n}[E[x_1]+\cdots+E[x_n]] = \frac{1}{n}\cdot n\mu = \mu.$$

$$\begin{aligned}Var(\bar{x}) &= var\left(\frac{x_1+\cdots+x_n}{n}\right)\\ &= \frac{1}{n^2}var(x_1+\cdots+x_n)\\ &= \frac{1}{n^2}[var(x_1)+\cdots+var(x_n)] \quad \cdots \text{ assuming the samples are independent.}\\ &= \frac{1}{n^2}\cdot n\cdot\sigma^2\end{aligned}$$

$$Var(\bar{x}) = \frac{\sigma^2}{n}$$

The expected value of sample mean is equal to the population mean and its variance is 1/n times the population variance.
The sample mean is also centered around the population mean μ, but its expected spread becomes more reduced as the sample size increases.

The Weak Law of Large Numbers

Let X1, X2, ... be a sequence of independent and identically distributed random variables, each having a finite mean E[Xi] = μ. Then for any ε ≥ 0,

$$P\left\{\left|\frac{x_1+x_2+\cdots+x_n}{n} - \mu\right| \geq \varepsilon\right\} \to 0 \quad \text{as } n \to \infty$$

Proof:

Sample mean $\bar{x} = \frac{x_1+\cdots+x_n}{n}$ $\quad E[\bar{x}] = \mu$, $var(\bar{x}) = \frac{\sigma^2}{n}$

$$P\{|\bar{x}-\mu| \geq \varepsilon\} = \frac{\left(\frac{\sigma^2}{n}\right)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \to \infty$,

$$\frac{\sigma^2}{n\varepsilon^2} \to 0$$

$$\therefore \lim_{n\to\infty} P\{|\bar{x}-\mu| \geq \varepsilon\} = 0$$

The Central Limit Theorem

Loosely speaking, it asserts that the sum of a large number of independent random variables has a distribution that is approximately normal.

Let X1, ..., Xn be a sequence of independent and identically distributed random variables with mean μ and variance σ².
Then for a sufficiently large n, the distribution of X1 + ... + Xn is approximately normal with mean nμ and variance nσ².

$$X_1 + \cdots + X_n \sim N(n\mu, n\sigma^2)$$

From the CLT,

$$\frac{X_1+\cdots+X_n - n\mu}{\sigma\sqrt{n}} \quad \text{is a standard normal rv approximately for a large } n.$$

$$P\left\{\frac{X_1+\cdots+X_n - n\mu}{\sqrt{n}\,\sigma} < x\right\} \approx P\{z < x\} = \phi(x)$$

Q. An insurance company has 25,000 policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and std deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

→ $X = X_1 + \cdots + X_{25000}$

$$P\{X_1 + \cdots + X_{25000} > 8.3 \times 10^6\}$$

$$= P\left\{\frac{X_1 + \cdots + X_{25000} - n\mu}{\sqrt{n}\,\sigma} > \frac{8.3 \times 10^6 - n\mu}{\sqrt{n}\,\sigma}\right\}$$

$$= P\left\{\frac{X - 25000 \cdot 320}{\sqrt{25000} \cdot 540} > \frac{8.3 \times 10^6 - 25000 \times 320}{\sqrt{25000} \times 540}\right\}$$

$$= P\{Z > 3.51\}$$

$$= 1 - P\{Z \le 3.51\}$$

$$= 1 - \Phi(0.351)$$

$$= 0.00023$$

Q. An instructor has 50 exams that'll be graded in sequence. The time required to grade them are independent with a common distribution that has mean 20 minutes and standard deviation 4 minutes. Approximate the probability that the instructor will grade atleast 25 exams in the first 450 minutes of work.

→ $X = X_1 + \cdots + X_{25}$ (X: time taken to grade 25 papers; X_i: time of ith paper)

$\mu_{X_i} = 20$, $\sigma_{X_i} = 4$

$$P\{X < 450\} = P\left\{\frac{X - n\mu}{\sqrt{n}\,\sigma} < \frac{450 - 25 \times 20}{\sqrt{25} \cdot 4}\right\}$$

$$= P\{Z < -2.5\}$$

Q. **EXAMPLE 6.3b** Civil engineers believe that W, the amount of weight (in units of 1,000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1,000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1? $\Phi(1.28) = 0.9$

→ $W \sim N(400, 40)$

$C_i \sim N(3, .3)$

$C = C_1 + C_2 + \cdots + C_n$ → approximately normal

$P\{C > W\}$ means prob. of structural damage

(C ≈ normal, W normal)

$= P\{C - W > 0\}$

↳ diff. of 2 normal variables is approx. normal

$E[C_1 + \cdots + C_n - W] = n\,E[C_i] - E[W] = 3n - 400$

$\text{var}(C_1 + \cdots + C_n - W) = n\,\text{var}(C_i) + \text{var}(W)$ $\qquad \text{var}(-X) = \text{var}(X)$

$= n(0.3)^2 + (40)^2$

$= 0.09n + 1600$

$$P\{C > W\} = P\{C - W > 0\}$$

$$= P\left\{\frac{C - W - E[C - W]}{\sigma} > 0\right\}$$

$$= P\left\{\frac{C - W - (3n - 400)}{\sqrt{0.09n + 1600}} > \frac{0 - (3n - 400)}{\sqrt{0.09n + 1600}}\right\}$$

$$= P\left\{Z > \frac{400 - 3n}{\sqrt{0.09n + 1600}}\right\} > 0.1$$

$$1 - \Phi\left(\frac{400 - 3n}{\sqrt{0.09n + 1600}}\right) > 0.1$$

$$\Phi\left(\frac{400 - 3n}{\sqrt{0.09n + 1600}}\right) < 0.9$$

$$\frac{400 - 3n}{\sqrt{0.09n + 1600}} < 1.28 \qquad \Phi(1.28) = 0.9$$

$$\Rightarrow n \ge 117$$

We can apply CLT on the binomial rv.

$X \sim \text{Binomial}(n, p)$

$X = X_1 + \cdots + X_n \qquad X_i \sim \text{Bernoulli}(p) \qquad X_i = \begin{cases} 1, & \text{if } i^{th} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$

If n is large, Binomial $\rightarrow$ Normal.

(CLT says so, as it is the sum of n independent rvs)

$E[X_i] = p \qquad \text{var}(X_i) = p(1-p)$

$E[X] = np \qquad \text{var}(X) = np(1-p)$

If n is large, X is approximately normal with mean np and var $np(1-p)$.

$\frac{X - np}{\sqrt{np(1-p)}}$ is approx. standard normal.

Continuity corrections

These are the adjustments that can be made to statistical tests when discontinuous distributions (like Binomial) are approximated to continuous ones (like Normal).

Discrete	Continuous
$x = c$	$c - 0.5 < x < c + 0.5$
$x < c$	$x < c + 0.5$
$x \leq c$	$x < c + 0.5$
$x > c$	$x > c - 0.5$
$x \geq c$	$x > c - 0.5$

Q. The ideal size of a 1st year class at a particular college is 150 students. The college, known from past experience that on average, 30% students who apply take admission, uses a policy of approving 450 applications. Compute the probability that more than 150 students attend the college.

→

of students joining, $X \sim \text{Binom}(450, 0.3)$

$P(\text{more than 150 students join}) = \sum_{i=151}^{450} \binom{450}{i} (0.3)^i (0.7)^{450-i}$ → actual prob. (too diff. to calculate)

Using CLT,

$X = X_1 + X_2 + \cdots + X_{450} \qquad X_i \sim \text{Bernoulli}(0.3)$

$E[X] = 450 \times 0.3 = 135$

$\text{Var}(X) = 450(0.3)(0.7) = 135 \times 0.7 = 94.5 \qquad \sigma_x = \sqrt{94.5} = 9.72$

$P\{X > 150\}$

↓ continuity correctn

$P\{X > 149.5\}$

$= P\left\{Z > \frac{149.5 - 135}{9.72}\right\} = P\{Z > 1.4917\}$

$= 1 - \Phi(1.4917)$

$= 1 - 0.9319$

$= 0.0681$

≈ 0.07

Approximate distribution of sample means

Let X1, X2, ..., Xn be a sample from a population having mean μ and variance σ^2. The CLT can be used to approximate the distribution of the sample mean.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Since $\sum_{i=1}^{n} x_i$ is approx. normal if n is large (CLT), $\bar{x}$ is also approx. normal if n is large with expected value μ and variance $\frac{\sigma^2}{n}$.

Also,

$$\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ is approx. std. normal if } n \text{ is large.}$$

Q. The weights of a population of workers have a mean of 167 and std dev 27.
a. If a sample of 36 workers is chosen. Approximate the probability that the sample mean of their weights lies between 163 and 170.
b. Repeat (a) when sample size is 144.

→ $\mu = 167, \ \sigma = 27$

a) $n = 36$

assume $\bar{x}$ is approx. normal.

$P\{163 < \bar{x} < 170\}$

$E[\bar{x}] = \mu = 167$

$\text{var}(\bar{x}) = \frac{\sigma^2}{n} = \frac{27^2}{36} = \frac{3}{4} \cdot 27 = \frac{81}{4}$

$\text{std dev}(\bar{x}) = \sigma_{\bar{x}} = \frac{9}{2} = 4.5$

$\frac{\bar{x} - 167}{9/2} \sim \text{Normal.}$

$P\{163 < \bar{x} < 170\}$

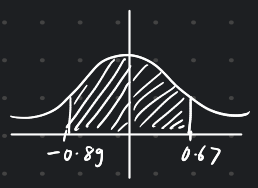
$= P\left\{\frac{163 - 167}{9/2} < \frac{\bar{x} - 167}{9/2} < \frac{170 - 167}{9/2}\right\}$

$= P\left\{\frac{-8}{9} < z < \frac{6}{9}\right\}$

$= P\{-0.89 < z < 0.67\}$

$= \varphi(0.67) + \varphi(0.89) - 1$

$= 0.5619$



b) $n = 144$

$\sigma_{\bar{x}}^2 = \frac{27^2}{144} = \frac{3}{16} \cdot 27 = \frac{81}{16} \quad \Rightarrow \sigma_{\bar{x}} = \frac{9}{4}$

$P\{163 < \bar{x} < 170\}$

$= P\left\{\frac{163 - 167}{9/4} < z < \frac{170 - 167}{9/4}\right\}$

$= P\left\{\frac{-16}{9} < z < \frac{12}{9}\right\}$

$= P\{-1.78 < z < 1.33\}$

$= \varphi(1.78) + \varphi(1.33) - 1$

$= 0.8707$

EXAMPLE 6.3e An astronomer wants to measure the distance from her observatory to a distant star. However, due to atmospheric disturbances, any measurement will not yield the exact distance d. As a result, the astronomer has decided to make a series of measurements and then use their average value as an estimate of the actual distance. If the astronomer believes that the values of the successive measurements are independent random variables with a mean of d light years and a standard deviation of 2 light years, how many measurements need she make to be at least 95 percent certain that her estimate is accurate to within $\pm .5$ light years? (given $\varphi(1.96) = 0.975$)

→ $\mu = d$, $\sigma = 2$

$n = ?$

$P\{d - 0.5 \le \bar{x} \le d + 0.5\} \ge 0.95$

$P\left\{\frac{d-0.5-d}{\frac{2}{\sqrt{n}}} < z < \frac{d+0.5-d}{\frac{2}{\sqrt{n}}}\right\} \ge 0.95$

$P\left\{-\frac{\sqrt{n}}{4} < z < \frac{\sqrt{n}}{4}\right\} \ge 0.95$

$\varphi\left(\frac{\sqrt{n}}{4}\right) - \varphi\left(-\frac{\sqrt{n}}{4}\right) \ge 0.95$

$2\varphi\left(\frac{\sqrt{n}}{4}\right) - 1 \ge 0.95$

$\varphi\left(\frac{\sqrt{n}}{4}\right) \ge 0.975$

$\varphi\left(\frac{\sqrt{n}}{4}\right) \ge \varphi(1.96)$

$\frac{\sqrt{n}}{4} \ge 1.96$

$n \ge 61.4656$

$n \ge 62$ at least 62 observations.

Q. If 10 fair dice are rolled, find the approximate probability that the sum obtained is between 30 and 40 (both inclusive).

→ X_i: result on i^{th} die $\sim$ Discrete uniform$(1, 6)$ $E[X] = \frac{a+b}{2} = 3.5$ $\text{Var}(X) = \frac{(b-a+1)^2 - 1}{12} = \frac{35}{12}$

(1 2 3 .. 6 ; $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$)

$X = X_1 + X_2 + \dots + X_n$ is approx. normal (CLT).

with $\bar{x} = 3.5 \times n = 35$

$\sigma_{\bar{x}}^2 = n \cdot \sigma^2 = \frac{350}{12}$

$\sigma_{\bar{x}} = \sqrt{\frac{350}{12}} \approx 5.4$

$P\{30 \le \bar{x} \le 40\}$

(continuity correc^n)

$P\{29.5 \le \bar{x} \le 40.5\}$

$= P\left\{\frac{29.5-35}{5.4} \le z \le \frac{40.5-35}{5.4}\right\}$

$= P\{-1.018 \le z \le 1.018\}$

$= 2\varphi(1.018) - 1$

$\approx 2\varphi(1.02) - 1$

$= 2(0.8461) - 1$

$= 0.692$

Q. The # of students who enroll in a psychology course is a Poisson rv with mean 100. What is the probability that more than 120 students will enroll?

→ $X \sim \text{Pois}(\lambda = 100)$

$E[X] = \lambda = 100$

$\text{Var}(X) = \lambda = 100$

$P\{X \geq 120\} = \sum_{i=121}^{\infty} \frac{e^{-100}(100)^i}{i!} = 1 - \sum_{i=0}^{120} \frac{e^{-100}(100)^i}{i!}$ diff. to calculate

Sum of 2 Poisson rv → Poisson with $\lambda = \lambda_1 + \lambda_2$

X = sum of 100 independent Poisson r.v. with $(\lambda = 1)$ = Poisson with $(\lambda = 100)$

So $X \sim$ Normal (using CLT)

$\bar{x} = 100$

$\sigma^2 = 100$, $\sigma = 10$

$P\{x > 120\}$

(continuity correction.

$P\{x > 119.5\}$

$= P\left\{\frac{x - 100}{10} > \frac{119.5 - 100}{10}\right\}$

$= P\{z > 1.95\}$

$= 1 - \varphi(1.95)$

$= 1 - 0.9744$

$= 0.0256$

The strong law of large numbers

It is the best known result in probability theory.
It states that the average of a sequence of independent rvs having a common distribution converges to the mean of the distribution with probability 1.

Theorem:
Let X1, X2, ..., be a sequence of IID random variables each having E[Xi] = μ, then with probability 1,

$$\frac{X_1 + X_2 + \dots + X_n}{n} \to \mu \quad \text{as} \quad n \to \infty$$

Suppose that a sequence of independent trials of some experiment is performed. Let E be a fixed event in the experiment. We define an indicator rv,

$$X_i = \begin{cases} 1, & \text{if } E \text{ happens in } i^{th} \text{ trial} \\ 0, & \text{otherwise.} \end{cases}$$

By the strong law of large #s,

$$\left\{\frac{X_1 + \dots + X_n}{n}\right\} \to \mu = E[X_i] = 1 \cdot P(E) + 0 \cdot P(\bar{E}) = P(E)$$

$$\frac{\#\text{ of times } E \text{ happens}}{\text{total \# of trials}} \to P(E) \text{ if \# of trials is large.}$$

One-sided Chebyshev inequality / Cantelli's inequality

$P(X-\mu \geq a) \leq \frac{\sigma^2}{\sigma^2+a^2}$

If X is a random variable with mean 0 and variance σ^2, then for any a>0,

$$P\{X \geq a\} \leq \frac{\sigma^2}{\sigma^2+a^2}$$

In probability theory, **Cantelli's inequality** (also called the **Chebyshev-Cantelli inequality** and the **one-sided Chebyshev inequality**) is an improved version of Chebyshev's inequality for one-sided tail bounds.[1] [2] [3] The inequality states that, for $\lambda > 0$,

$$\Pr(X - \mathbb{E}[X] \geq \lambda) \leq \frac{\sigma^2}{\sigma^2 + \lambda^2},$$

$P(X-\mu \geq a) \leq \frac{\sigma^2}{\sigma^2+a^2}$

Attempted solution:

I know the Chebyshev's inequality which States that

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

If I first argue that for any $b > 0$

$$P(X \geq a) \leq P[(X+b)^2 \geq (a+b)^2]$$
$$\implies P(X \geq a) \leq \frac{E(X+b)^2}{(a+b)^2}$$
$$P(X \geq a) \leq \frac{E(X^2)+2E(X)b+b^2}{(a+b)^2}$$
$$P(X \geq a) \leq \frac{\sigma^2+b^2}{(a+b)^2}$$

Proof:

$$P\{X \geq a\} = P\{X+b \geq a+b\}$$
$$= P\{(X+b)^2 \geq (a+b)^2\}$$

$(X+b)^2$ is a non-negative rv.

Using markov's inequality,

$$P\{X \geq a\} \leq P\{(X+b)^2 \geq (a+b)^2\}$$
$$\leq \frac{E[(X+b)^2]}{(a+b)^2}$$
$$\leq \frac{E[X^2+2Xb+b^2]}{(a+b)^2}$$
$$\leq \frac{E[X^2]+2bE[X]+E[b^2]}{(a+b)^2} \qquad E[X]=0$$
$$\leq \frac{\mathrm{var}(X)+(E[X])^2+0+b^2}{(a+b)^2}$$

$$P\{X \geq a\} \leq \frac{\sigma^2+b^2}{(a+b)^2}, \quad b \geq 0$$

we'd minimize the RHS to get the better bound.

we get minima at $b = \frac{\sigma^2}{a}$

$$P\{X \geq a\} \leq \frac{\sigma^2+\frac{\sigma^4}{a^2}}{(a+\frac{\sigma^2}{a})^2}$$
$$\leq \frac{\sigma^2(\sigma^2+a^2)}{(\sigma^2+a^2)^2}$$
$$P\{X \geq a\} \leq \frac{\sigma^2}{\sigma^2+a^2}$$

□

$\frac{d}{db}\left(\frac{\sigma^2+b^2}{(a+b)^2}\right) = 0$ $\qquad \left(\frac{u}{v}\right)' = \frac{vu'-uv'}{v^2}$

$$\frac{(a+b)^2(2b)-(\sigma^2+b^2)(2(a+b))}{(a+b)^4} = 0$$
$$(a+b)\left[(a+b)b-(\sigma^2+b^2)\right] = 0$$
$$b^2+ab-\sigma^2-b^2 = 0$$
$$b = \frac{\sigma^2}{a}$$

Q. If the # of items produced in a factory during a week is a random variable with mean 100 and variance 400, compute an upper bound on the probability that this week's production will be at least 120.

→ $E(X) = 100$

$\text{var}(X) = 400$

Using markov inequality, $P\{X \geq 120\} \leq \frac{E(X)}{120} = \frac{100}{120}$

$$\boxed{P\{X \geq 120\} \leq \frac{5}{6}}$$

Using one sided chebyshev ineq $P\{X - \mu \geq a\} \leq \frac{\sigma^2}{\sigma^2 + a^2}$

$$P\{X - 100 \geq \underbrace{120 - 100}_{a=20}\} \leq \frac{400}{400 + 20^2}$$

$$\boxed{P\{X \geq 120\} \leq \frac{1}{2}} \longrightarrow \text{Better bound.}$$

The sample variance

Let X1, ..., Xn be a random sample i.i.d. from a population with mean μ and variance σ^2.

let $\bar{x} = \frac{\sum x_i}{n}$ be the sample mean.

Sample variance: $S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$ S is the sample std. dev^n.

$$(n-1)S^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$E[(n-1)S^2] = E\left[\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right]$$

$$= \sum_{i=1}^{n} E[x_i^2] - n\,E[\bar{x}^2]$$

$$= n\,E[x_i^2] - n\,E[\bar{x}^2]$$

$$= n\left[\left[\text{var}(x_i) + E[x_i]^2\right] - \left[\text{var}(\bar{x}) + E[\bar{x}]^2\right]\right]$$

$$= n\left[\sigma^2 + \mu^2 - \frac{\sigma^2}{n} - \mu^2\right]$$

$$= n \cdot \sigma^2\left(\frac{n-1}{n}\right)$$

$$(n-1)\,E[S^2] = (n-1)\,\sigma^2$$

$$E[S^2] = \sigma^2$$

$$\sum_{i=1}^{n} (x_i - \bar{x})^2$$
$$= \sum_{i=1}^{n} (x_i^2 - 2x_i\bar{x} + \bar{x})^2$$
$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + \bar{x}\sum_{i=1}^{n} 1$$
$$= \sum_{i=1}^{n} x_i^2 - 2n\bar{x}\cdot\bar{x} + \bar{x}^2 \cdot n$$
$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

The expected value of sample variance is popula^n variance.

Sampling from a normal population

Theorem:
If x1, ..., xn is a sample from a normal population (μ, σ^2),

then $\bar{x}$ and s^2 are independent random variables,

with $\bar{x}$ being normal $(\mu, \frac{\sigma^2}{n})$.

and $(n-1)\frac{s^2}{\sigma^2}$ being a chi-squared distribu^n with (n-1) degrees of distribu^n.

Proof:

Popula^n ~ $N(\mu, \sigma^2)$

↳ Sample: $x_1, x_2, \ldots, x_n \sim N$ $\quad E[x_i] = \mu,\ \text{var}(x_i) = \sigma^2$

$\bar{x} = \frac{\sum x_i}{n}$ $\quad E[\bar{x}] = \mu,\ \text{var}[\bar{x}] = \frac{\sigma^2}{n}$

$s^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)}$ $\quad E[S^2] = \sigma^2$

① Distribu^n of sample mean:

Since $x_1, \ldots, x_n$ are normal,

$\bar{x} = \frac{x_1 + \cdots + x_n}{n}$ is normal

$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$

② Distribu^n of sample variance:

$S = \frac{\sum (x_i - \bar{x})^2}{n-1} \quad \Rightarrow \quad \frac{(n-1)s^2}{\sigma^2} = \frac{\sum (x_i - \bar{x})^2}{\sigma^2}$

$x_1 \ldots x_n$: $\bar{x} = \frac{x_1 + \cdots + x_n}{n}$ $\quad y_i = x_i - \mu$ ↷ popl^n mean

y_i's: $(x_1 - \mu), (x_2 - \mu) \cdots$

$$\bar{y} = \frac{(x_1 - \mu) + (x_2 - \mu) + \cdots + (x_n - \mu)}{n}$$

$$\bar{y} = \frac{x_1 + \cdots + x_n}{n} - \frac{n\mu}{n} = \bar{x} - \mu$$

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2 \quad \ldots \text{ proved earlier}$$

$$\sum_{i=1}^{n}(x_i-\mu-(\bar{x}-\mu)) = \sum(x_i-\mu)^2 - n(\bar{x}-\mu)^2$$

$$\sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n}(x_i-\mu)^2 - n(\bar{x}-\mu)^2$$

$$\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^2} - \frac{n(\bar{x}-\mu)^2}{\sigma^2}$$

$$= \sum_{i=1}^{n}\underbrace{\left(\frac{x_i-\mu}{\sigma}\right)^2}_{\text{std. normal}} - \underbrace{\left(\frac{\bar{x}-\mu}{\frac{\sigma^2}{n}}\right)^2}_{\text{std. normal}}$$

$$= \chi^2_{(n-1)}$$

so if pop^n is normal, sample variance follows the χ^2 distribution

---

Let x1, ..., xn be a sample from a normal population (μ, σ^2). If $\bar{x}$ denotes the sample mean, s^2 denotes the sample variance, then

$$\frac{\sqrt{n}(\bar{x}-\mu)}{s} \sim t_{n-1}$$ (t-dist with n-1 degrees of freedom)

Point estimation

It means to estimate different values related to the population using the sample.
E.g. Population ~ N(μ, σ^2). We sample 10 points. We want to estimate the mean and variance of the population.

Point estimation:
Estimating the parameters of a population using a sample from the population.
Assume that θ is an unknown parameter to be estimated. To estimate θ, we need to collect some data.
Specifically, we get a random sample x1, ..., xn such that all these xi's have the same distribution as the population.
To estimate θ, we define a point estimator, that is a function of the random sample.

$$\hat{\theta} = h(x_1, x_2, \ldots, x_n)$$

E.g. $\theta = E[X]$ (pop^n)

h : avg. of all values in a sample.

$\hat{\theta} = \frac{x_1 + \ldots + x_n}{n}$

$\hat{\theta}$ (our estimate) should be a good estimate of θ (actual value). $\theta - \hat{\theta} = 0$

Evaluating estimators

Bias:
The bias of an estimator tells us on average how far the estimator is from the actual value.
($\hat{\theta}$) (θ)

Let $\hat{\theta} = h(x_1, \ldots, x_n)$ be a point estimator of θ.

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Unbiased estimator : $B(\hat{\theta}) = 0$

Q. Let $x_1, \ldots, x_n$ be a random sample. Show that the sample mean is an unbiased estimator of pop^n mean.

→ $\theta = E[X] = \mu$

$\bar{x} = \text{sample mean} = \frac{x_1 + \ldots + x_n}{n}$ (point estimator $\hat{\theta}$)

$B(\hat{\theta}) = E[\hat{\theta}] - \theta$
$= E[\bar{x}] - \mu$
$= \mu - \mu$
$= 0$

∴ The sample mean is an unbiased estimator of the population mean.

Q. Let $x_1, \ldots, x_n$ be a sample from a pop^n. Let the point estimator be $\hat{\theta} = x_1$ for population mean. Calculate bias of $\hat{\theta}$.

→ $\hat{\theta} = x_1$

$B(\hat{\theta}) = E[\hat{\theta}] - \theta$
$= E[x_1] - \mu$
$= \mu - \mu = 0$

Both $\hat{\theta} = x_1$ and $\hat{\theta} = \bar{x}$ are unbiased.

If an estimator is unbiased, it doesn't necessarily mean that it's a good estimator.

So bias is not really a good measure of how good an estimator is.

Mean squared error:

The mean squared error of a point estimator is defined as

$$MSE(\hat{\theta}) = E[(\hat{\theta}-\theta)^2]$$

Q. Let $x_1, \ldots, x_n$ be a random sample from a distribⁿ with mean $E[x] = \theta$ and variance $var(x) = \sigma^2$.
Calculate the MSE for

a) $\hat{\theta} = x_1$

b) $\hat{\theta} = \frac{x_1 + \ldots + x_n}{n}$

→ a) $\hat{\theta}_1 = x_1$

$E[(\hat{\theta}_1 - \theta)^2] = E[(x_1 - \mu)^2]$

$= E[(x_1 - E[x_1])^2]$

$= var(x_1)$

$MSE[\hat{\theta}_1] = \sigma^2$

b) $\hat{\theta}_2 = \frac{x_1 + \ldots + x_n}{n} = \bar{x}$

$E[(\hat{\theta}_2 - \hat{\theta})^2] = E[(\bar{x} - \mu)^2]$

$= E[(\bar{x} - E[\bar{x}])^2]$

$= var(\bar{x})$

$MSE[\hat{\theta}_2] = \frac{\sigma^2}{n}$

If $n=1$, $MSE[\hat{\theta}_1] = MSE[\hat{\theta}_2]$

For $n>1$, $MSE[\hat{\theta}_1] > MSE[\hat{\theta}_2]$

$\hat{\theta}_2$ is a better estimator of θ.

In all of this, θ is treated as an unknown constant.

$$MSE = E[(\hat{\theta}-\theta)^2]$$

$$= Var(\hat{\theta}-\theta) + E[(\hat{\theta}-\theta)]^2 \quad \text{as } E[x^2] = var[x] + E[x]^2$$

$$= Var(\hat{\theta}) + E[\hat{\theta}-\theta]^2$$

$$MSE_{\hat{\theta}} = Var(\hat{\theta}) + Bias(\hat{\theta})^2$$

Consistency:

Let $\hat{\theta}_n$ be a point estimator for θ defined over a sample of size n. Then $\hat{\theta}_n$ is said to be a consistent estimator of θ iff

$$\lim_{n\to\infty} P[|\hat{\theta}_n - \theta| \geq \varepsilon] = 0 \qquad \forall\ \varepsilon > 0.$$

} meaning for a really large sample, $\hat{\theta} = \theta$

Q. Let $x_1, \ldots, x_n$ be a random sample from a distribⁿ with mean $E[x] = \theta$ and variance $var(x) = \sigma^2$.
Check if these are consistent estimators of θ.

a) $\hat{\theta} = x_1$

b) $\hat{\theta} = \frac{x_1 + \ldots + x_n}{n}$

→ a) $\lim_{n\to\infty} P\{|x_1 - \mu| \geq \varepsilon\} \neq 0$ } $\hat{\theta} = x_1$ is an inconsistent estimator of θ.

b) $\lim_{n\to\infty} P\{|\bar{x} - \mu| \geq \varepsilon\} = 0$ → By the weak law of large #s.

$\hat{\theta} = \bar{x}$ is a consistent estimator for θ.

Point estimator for variance

Q: Let $x_1, \dots, x_n$ be a random sample. Find the bias of $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$ to estimate the popula^n variance.

→ $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$

$= E\left[\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right] - \theta$

$= E\left[\frac{1}{n}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)\right] - \sigma^2$

↳ proved previously

$= \frac{1}{n}\left[\sum_{i=1}^{n} E[x_i^2] - n\,E[\bar{x}^2]\right] - \sigma^2$

$= \frac{1}{n}\left[\sum_{i=1}^{n}\left[\text{var}(x_i) + E[x_i]^2\right] - n\left(\text{var}(\bar{x}) + E[\bar{x}]^2\right)\right] - \sigma^2$

$= \frac{1}{n}\left[n\sigma^2 + n\mu^2 - n\left[\frac{\sigma^2}{n} + \mu^2\right]\right] - \sigma^2$

$= \left(\frac{n-1}{n}\right)\sigma^2 - \sigma^2$

$B[\hat{\theta}] = -\frac{\sigma^2}{n}$ → if it were $\frac{n-1}{n-1}$, Bias would be 0.

↳ Biased

That's why the $(n-1)$.

The sample variance $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ is an unbiased estimator of the population variance.

How do we do this for some other parameter?

Maximum likelihood estimation

Q. I have a bag that contains 3 balls. Each ball is either red or blue. That's all the information I have.
The # of blue balls, call it θ, might be 0, 1, 2, or 3.
We sample (with replacement) 4 balls from this bag.
We define random variables X1, X2, X3, X4 as follows:

$$X_i = \begin{cases} 1, & \text{if the } i^{th} \text{ ball is blue} \\ 0, & \text{otherwise} \end{cases}$$

Note that Xi's are i.i.d. random variables.
Xi ~ Bernoulli (p = θ/3)

After doing the experiment, we get the values:
X1=1, X2=0, X3=1, X4=1 (3 blue, 1 red)

Estimate the value of θ.

→

$X_i \sim \text{Bernoulli}(\theta/3)$ $\quad P_{x_i}(x_i) = \begin{cases} \theta/3, & x_i = 1 \\ 1-\theta/3, & x_i = 0 \end{cases}$

we calculate joint prob of X_1, X_2, X_3, X_4.

$P_{X_1,X_2,X_3,X_4}(x_1,x_2,x_3,x_4) = P(X_1=x_1 \text{ and } X_2=x_2 \text{ and } X_3=x_3 \text{ and } X_4=x_4)$
$= P_{X_1}(x_1) \cdot P_{X_2}(x_2) \cdot P_{X_3}(x_3) \cdot P_{X_4}(x_4)$ .. as they're independent (sampling w/ replacement)

$P_{X_1,X_2,X_3,X_4}(1,0,1,1) = P_{X_1}(1) \cdot P_{X_2}(0) \cdot P_{X_3}(1) \cdot P_{X_4}(1)$
$= \frac{\theta}{3} \cdot \left(1-\frac{\theta}{3}\right) \cdot \frac{\theta}{3} \cdot \frac{\theta}{3}$

$= \left(\frac{\theta}{3}\right)^3 \left(1-\frac{\theta}{3}\right)$ → likelihood function (probability of obtaining that sample)

We need a θ that maximizes the likelihood func^n.
we could solve it as an optimiza^n problem.
But $\theta = \{0, 1, 2, 3\}$. So just check individually. ↳ we'd do that when there are too many values of n to check.

$P_{X_1X_2X_3X_4}(1,0,1,1;\theta=0) = \left(\frac{0}{3}\right)^3\left(1-\frac{0}{3}\right) = 0$
$P_{X_1X_2X_3X_4}(1,0,1,1;\theta=1) = \left(\frac{1}{3}\right)^3\left(1-\frac{1}{3}\right) = \frac{2}{81}$
$P_{X_1X_2X_3X_4}(1,0,1,1;\theta=2) = \left(\frac{2}{3}\right)^3\left(1-\frac{2}{3}\right) = 8/81$ → max. value of likelihood func^n.
$P_{X_1X_2X_3X_4}(1,0,1,1;\theta=3) = \left(\frac{3}{3}\right)^3\left(1-\frac{3}{3}\right) = 0$

$\hat{\theta} = 2$ maximizes the likelihood func^n, and gives the best estimate for θ.

Maximum likelihood estimation

Let $X_1, \cdots, X_n$ be a random sample from a distribution with parameter θ. Suppose $x_1, x_2, \ldots, x_n$ are the observed values of the sample $X_1, \ldots, X_n$.
If X_i's are discrete RVs, we define the likelihood as the prob. of the observed sample as a func^n of θ.

$$L(x_1, \ldots, x_n ; \theta) = P(X_1 = x_1, \ldots, X_n = x_n)$$
$$= P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n ; \theta)$$

$$L(\mathbf{x} ; \theta) = P_{\mathbf{x}}(\mathbf{x} ; \theta)$$
↳ vector

If $X_1, \ldots, X_n$ are continuous, then we use the joint PDF in stead of the joint PMF.

$$L(x_1, \ldots, x_n ; \theta) = f_{X_1, \ldots, X_n}(x_1, \ldots, x_n ; \theta)$$

NOTE: often, log(likelihood) is simpler to optimize.

The maximum likelihood estimate of θ, shown by $\hat{\theta}_{ML}$ is the value that maximizes the likelihood function $L(x_1, x_2, \ldots, x_n ; \theta)$
In determining the max. value of the likelihood, it is often useful to use the fact that $f(x_1, x_2, \ldots, x_n ; \theta)$ and $\log[f(x_1, \ldots, x_n ; \theta)]$ have the maximum values at the same $\hat{\theta}$.

Maximum likelihood estimator for the Bernoulli parameter

Suppose that n independent Bernoulli trials (p) are performed. The observed values of trials are x1, ..., xn. What is the maximum likelihood estimator for p?

→ $X_i = \begin{cases} 1, & i^{th} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$

PMF X_i : $P_{x_i}(x_i) = \begin{cases} p & \text{if } x_i = 1 \\ (1-p) & \text{if } x_i = 0 \end{cases}$

$$P\{X_i = x_i\} = p^{x_i} \cdot (1-p)^{1-x_i}$$

$$L(x_1, \ldots, x_n \mid p) = P_{x_1 \ldots x_n}(x_1, \ldots, x_n \mid p)$$
$$= P_{x_1}(x_1) \cdots P_{x_n}(x_n)$$
$$= p^{x_1}(1-p)^{1-x_1} \ldots p^{x_n}(1-p)^{1-x_n}$$

$$L(x_1 \ldots x_n \mid p) = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

$$\text{log likelihood} = \log L(x_1 \ldots x_n \mid p)$$
$$= \sum_{i=1}^{n} x_i \log p + (n - \sum_{i=1}^{n} x_i) \log(1-p)$$

$$\frac{d}{dp}(\text{log likelihood}) = 0$$

$$\frac{\sum x_i}{\hat{p}} + \frac{(n - \sum x_i)(-1)}{(1-\hat{p})} = 0$$

$$\sum x_i - \hat{p} \sum x_i - n\hat{p} + \hat{p} \sum x_i = 0$$

$$\boxed{\hat{p} = \frac{\sum x_i}{n}}$$

Best estimate for prob of success = $\frac{\#\text{ of successes}}{\#\text{ of trials}}$

Maximum likelihood estimator for the Binomial parameter

Q. $X_i \sim \text{Binomial}(3, \theta)$ and the observed values of X_i are $(x_1, x_2, x_3, x_4) = (1, 3, 2, 2)$. What is the max. likelihood estimate of θ?

→ $P_{X_i}(x_i) = \binom{3}{x_i} \theta^{x_i} (1-\theta)^{3-x_i}$

$$L(x_1 x_2 x_3 x_4 \mid \theta) = P_{X_1 X_2 X_3 X_4}(1, 3, 2, 2 \mid \theta)$$

$$= P_{X_1}(1) \cdot P_{X_2}(3) \cdot P_{X_3}(2) \cdot P_{X_4}(2)$$

$$= \binom{3}{1} \theta^1 (1-\theta)^2 \binom{3}{3} \theta^3 (1-\theta)^0 \cdot \binom{3}{2} \theta^2 (1-\theta)^1 \cdot \binom{3}{2} \theta^2 (1-\theta)^1$$

$$= 27\, \theta^8 (1-\theta)^4$$

$$\log L(\vec{X} \mid \theta) = \log 27 + 8 \log \theta + 4 \log(1-\theta)$$

$$\frac{d}{d\theta} \log L = 0$$

$$\frac{8}{\theta} - \frac{4}{1-\theta} = 0$$

$$\frac{2}{\theta} = \frac{1}{1-\theta}$$

$$2 - 2\theta = \theta$$

$$\hat{\theta} = 2/3$$

For the generic case,

Q. $X_i \sim \text{Binom}(m, \theta)$. MLE of θ?

→ $P_{X_i}(x_i) = \binom{m}{x_i} \theta^{x_i} (1-\theta)^{(m-x_i)}$

$$\hat{\theta} = \frac{\sum_{i=1}^{n} x_i}{m \cdot n}$$

Maximum likelihood estimator for the Exponential parameter

Q. $X_i \sim Exp(\theta)$. observed values of the sample: $(x_1, x_2, x_3, x_4) = (1.23, 3.32, 1.98, 2.12)$

MLE $(\theta) = ?$

→ $P_{X_i}(x_i) = \theta e^{-\theta x_i}, \quad x_i \geq 0$

$L(\vec{X}|\theta) = f_{X_1 X_2 X_3 X_4}(x_1, x_2, x_3, x_4 | \theta)$

$= \theta e^{-\theta x_1} \cdot \theta e^{-\theta x_2} \cdot \theta e^{-\theta x_3} \cdot \theta e^{-\theta x_4} = \theta e^{-(\theta_1 + \theta_2 + \theta_3 + \theta_4)}$

$= \theta^4 e^{-\theta(8.65)}$

$\log L(\vec{x}|\theta) = 4 \log \theta - 8.65\theta$

$\frac{d}{d\theta} \log L(\vec{x}|\theta) = \frac{4}{\hat{\theta}} - 8.65 = 0$

$\hat{\theta} = 0.46$

For n values,

$$\hat{\theta} = \frac{n}{\text{sum of observed values}}$$

$X_i \sim Exp(\theta)$ and we observe a sample $(x_1, x_2, \cdots, x_n) = (x_1, \cdots, x_n)$.

MLE for θ is

$$\hat{\theta} = \frac{n}{\sum_{i=1}^{n} x_i}$$

Maximum likelihood estimator for the uniform parameter

Let $X_1, \dots, X_n$ be a random sample from $U_{[0,\theta]}$, where θ is unknown.
We observe a sample and get values $(x_1, \dots, x_n)$.

$$P_{X_i}(x_i) = \frac{1}{\theta - 0} = \frac{1}{\theta}$$

$$\begin{aligned} L(\bar{x} \mid \theta) &= P_{X_1 \dots X_n}(x_1 \dots x_n) \\ &= P_{X_1}(x_1) \cdots P_{X_n}(x_n) \\ &= \begin{cases} \frac{1}{\theta} \cdots \frac{1}{\theta} & \text{if } 0 \le x_i \le \theta \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{\theta^n} & \text{if } 0 \le x_i \le \theta \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

To maximize $\frac{1}{\theta^n}$, we need the smallest θ possible. [$\theta = 0$ would give ∞, we'll not consider that]

$\hat{\theta}_{ML} = \max(x_1, \dots, x_n)$ as θ is anyways $\ge \max(x_1, \dots, x_n)$

Similarly for $U_{[a,b]}$, $MLE(a) = \min(x_1, \dots, x_n)$
$MLE(b) = \max(x_1, \dots, x_n)$

Maximum likelihood estimator for the Poisson parameter

Say $X_1, \dots, X_n$ are independent Poisson(λ) RVs with observed values $(x_1, \dots, x_n)$. MLE(λ) = ?

$$P_{X_i}(x_i) = e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \qquad x_i = 0, 1, 2, \dots$$

$$L(\vec{x} \mid \lambda) = e^{-\lambda} \frac{\lambda^{x_1}}{x_1!} \cdots e^{-\lambda} \frac{\lambda^{x_n}}{x_n!}$$

$$= \frac{e^{-n\lambda} \cdot \lambda^{\sum x_i}}{\prod x_i!}$$

$$\log(L(\vec{x} \mid \lambda)) = -n\lambda + \sum x_i \log\lambda - \log(\prod x_i!)$$

$$\frac{d}{d\hat{\lambda}} \log L(\vec{x} \mid \hat{\lambda}) = -n + \frac{\sum x_i}{\hat{\lambda}} - 0 = 0$$

$$\hat{\lambda} = \frac{\sum x_i}{n}$$

Maximum likelihood estimator for the normal parameters

Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ are independent. We observe the sample values $(x_1, \dots, x_n)$.

Here θ is a vector $\begin{bmatrix} \mu \\ \sigma \end{bmatrix}$

$$f_{X_i}(x_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i-\mu}{\sigma}\right)^2} \qquad x_i \in \mathbb{R}$$

$$L(\vec{x} \mid (\mu,\sigma)) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$$

$$= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2}$$

$$\log L(\vec{x} \mid \mu,\sigma) = -n\log(\sigma) + \log(\sqrt{2\pi}) - \frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2$$

$$\frac{\partial \log L}{\partial \mu} = -\frac{1}{2}\sum_{i=1}^{n}(-2)\left(\frac{x_i-\hat{\mu}}{\sigma^2}\right)$$

$$= \sum_{i=1}^{n}\left(\frac{x_i-\hat{\mu}}{\sigma^2}\right) = 0$$

$$\sum_{i=1}^{n} x_i - n\hat{\mu} = 0$$

$$\hat{\mu} = \frac{\sum x_i}{n}$$

$$\frac{\partial \log L}{\partial \sigma} = -\frac{n}{\sigma} - \frac{1}{2}\left(-2\sum(x_i-\mu)^2\sigma^{-3}\right)$$

$$= -\frac{n}{\sigma} + \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^3}$$

$$\frac{\sum_{i=1}^{n}(x_i-\hat{\mu})^2}{n} = \hat{\sigma}^2$$

$$\hat{\sigma} = \sqrt{\frac{\sum(x_i-\hat{\mu})^2}{n}}$$

Interval estimation

Instead of estimating the value of the unknown parameter θ, we'll now report an interval $[\hat{\theta}_l, \hat{\theta}_h]$ which we hope includes the real value of θ.

- length of interval $= \hat{\theta}_h - \hat{\theta}_l$
  - It shows the precision with which we can estimate θ. The smaller the interval, the higher the precision with which we estimate θ.

Confidence interval: shows how confident we are about the interval.

Let $X_1, \ldots, X_n$ be a random sample from a distribⁿ with a parameter θ to be estimated. Our goal is to find 2 estimators for θ.

1. The low estimator $\hat{\theta}_l = h_1(X_1, \ldots, X_n)$
2. The high estimator $\hat{\theta}_h = h_h(X_1, \ldots, X_n)$.

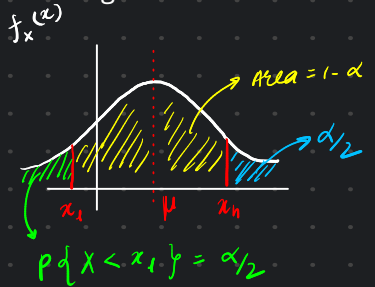
The estimators $\hat{\theta}_l$ and $\hat{\theta}_h$ are chosen s.t. the probability that $[\hat{\theta}_l, \hat{\theta}_h]$ contains θ is more than $(1-\alpha)$.

$$P\{\hat{\theta}_l \le \theta \le \hat{\theta}_h\} \ge 1-\alpha$$

common values of α: 0.1, 0.05, 0.01
for confidences: 90%, 95%, 99%

$[\hat{\theta}_l, \hat{\theta}_h]$ is a $(1-\alpha)\cdot 100\%$ confidence interval for θ.

Finding interval estimators



we report an interval centred about the mean.

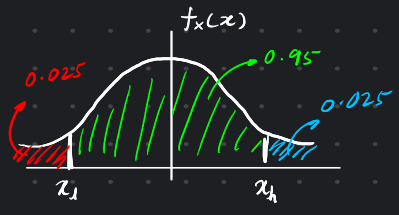
$$P\{X < x_l\} = \alpha/2 \qquad P\{X > x_h\} = \frac{\alpha}{2}$$

$$F_X(x_l) = \alpha/2 \qquad F_X(x_h) = 1 - \alpha/2$$

$$x_l = F_X^{-1}(\alpha/2) \qquad x_h = F_X^{-1}(1-\alpha/2)$$

Q. Let $Z \sim N(0,1)$. Find x_l & x_h s.t. $P(x_l \le Z \le x_h) = 0.95$.

→



$\alpha = 0.05$

$F_X(x_l) = 0.025$ $\qquad$ $F_X(x_h) = 0.975$

$x_l = F_X^{-1}(0.025)$ $\qquad$ $x_h = \varphi^{-1}(0.975)$

$= \varphi^{-1}(0.025)$ $\qquad$ $x_h = 1.96$

$x_l = -1.96$

$$P(-1.96 < Z < 1.96) = 0.95$$

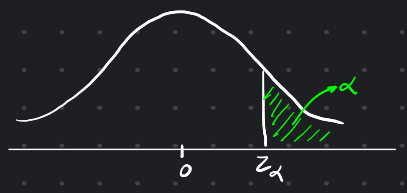
Z-score

for $q \in (0,1)$, let z_α be s.t.

$$P\{Z > z_\alpha\} = 1 - \varphi(z_\alpha) = \alpha$$

$$P(z < x) = 1 - \alpha$$

$$\varphi(x) = 1 - \alpha$$



e.g. $z_{0.05}$

$P(z > x) = 0.05$

$\Rightarrow P(z \le x) = 1 - 0.05 = 0.95$

$\varphi(x) = 0.95$

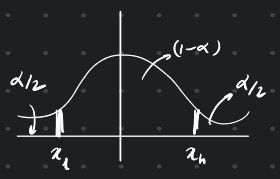
$x = \varphi^{-1}(0.95)$

$x = 1.645$

$z_{0.05} = 1.645$

Q. let $Z \sim N(0,1)$. Find x_1 & x_n s.t. $P(x_1 \le Z \le x_n) = 1-\alpha$

→



$P\{z > x_n\} = \alpha/2$

$1 - \varphi(x_n) = \alpha/2$

$x_n = \varphi^{-1}(1 - \alpha/2)$

$= z_{\alpha/2}$

$x_1 = -z_{\alpha/2}$ (By symmetry of std. normal curve)

$$P\{-z_{\alpha/2} < z < z_{\alpha/2}\} = 1 - \alpha$$

Finding interval $[\hat{\theta}_1, \hat{\theta}_n]$ such that $P\{\theta \in [\hat{\theta}_1, \hat{\theta}_n]\} = 1-\alpha$

e.g. Let $X_1, \dots X_n$ be a random sample from $N(\theta, 1)$. Find a 95% confidence interval for θ.

→ we start with a point estimator of θ.

$$\hat{\theta} = \frac{X_1 + \dots + X_n}{n}$$

Since X_i's are normal $(\theta, 1)$ rvs, $\hat{\theta}$ is also a normal rv. with mean μ and variance $1/n$.

$$\bar{x} \sim N(\theta, \tfrac{1}{n})$$

$\therefore \frac{\bar{x} - \theta}{1/\sqrt{n}}$ is a std. normal variable, $N(0,1)$

$$\sqrt{n}\,(\bar{x} - \theta) \sim N(0,1)$$

We know $P\{-z_{\alpha/2} \le z \le z_{\alpha/2}\} = 1-\alpha$

$P\{-z_{0.05} \le \sqrt{n}\,(\bar{x}-\theta) \le z_{0.05}\} = 0.95 = 1 - 0.05 \quad (\alpha = 0.05)$

$P\{-1.96 \le \sqrt{n}\,(\bar{x}-\theta) \le 1.96\} = 0.95$

$$P\left\{\frac{-1.96}{\sqrt{n}} - \bar{x} \le -\theta \le \frac{1.96}{\sqrt{n}} - \bar{x}\right\} = 0.95$$

$$P\left\{\bar{x} + \frac{1.96}{\sqrt{n}} > \theta > \bar{x} - \frac{1.96}{\sqrt{n}}\right\} = 0.95$$

$$P\left\{\theta \in \left[\bar{x} - \frac{1.96}{\sqrt{n}}, \bar{x} + \frac{1.96}{\sqrt{n}}\right]\right\} = 0.95$$

$$\left[\hat{\theta}_1, \hat{\theta}_n\right] = \left[\bar{x} - \frac{1.96}{\sqrt{n}}, \bar{x} + \frac{1.96}{\sqrt{n}}\right]$$

Let $X_1, \dots, X_n$ be a sample from a distribn with known variance $\text{Var}(X_i) = \sigma^2$, and unknown mean $E[X_i] = \theta$. Find a $(1-\alpha)$ conf. interval for θ. Assume that n is large.

→ Using CLT, $\bar{x}$ is approx. normal.

$$\bar{x} \sim N(\theta, \tfrac{\sigma^2}{n})$$

$$\Rightarrow \frac{\bar{x} - \theta}{\frac{\sigma}{\sqrt{n}}} \sim N[0,1]$$

$$P\left\{-z_{\alpha/2} < \frac{\bar{x}-\theta}{\sigma/\sqrt{n}} < z_{\alpha/2}\right\} = 1-\alpha$$

$$P\left\{\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \theta < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$$

$\left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$ is a $(1-\alpha)$ conf. interval for θ.

For 95%, $\left[\bar{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right]$

Q. When a signal having value μ is transmitted from a locn A. The value received at B is $N(\mu, 4)$. To reduce error, the same value is sent 9 times. The successive values received are $[5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5]$.
What is the 95% conf. interval for μ?

→ $\bar{x} = \frac{5 + 8.5 + \dots + 10.5}{9} = \frac{81}{9} = 9.$

$\sigma = 2$

conf. interval $= \left[\bar{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right] = \left[9 - 1.96\frac{2}{\sqrt{9}},\ 9 + 1.96\frac{2}{\sqrt{9}}\right] = [7.69,\ 10.31]$

confidence interval	α	$\alpha/2$	$z_{\alpha/2}$	Explⁿ
90%	0.1	0.05	1.65	$\varphi(1.65) \approx 0.95$
95%	0.05	0.025	1.96	$\varphi(1.96) \approx 0.975$
99%	0.01	0.005	2.58	$\varphi(2.58) \approx 0.995$

Q. An engineer is measuring a quantity θ. It is assumed that there is a random error in each measurement, so the engineer will take n measurements and report the average of the measurements as the estimated value of θ. Here, n is assumed to be large enough so that the central limit theorem applies. If Xi is the value that is obtained in the ith measurement, we assume that

$$X_i = \theta + W_i$$

where Wi is the error in the ith measurement. We assume that the Wis are i.i.d. with E[Wi]=0 and Var(Wi)=4 square units. The engineer reports the average of the measurements

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

How many measurements does the engineer need to make until he is 90% sure that the final error is less than 0.25 units? In other words, what should the value of n be such that

$$P(\theta - 0.25 \le \bar{X} \le \theta + 0.25) \ge 0.90 ?$$

→ Error = W_i

$W_1, W_2, \ldots, W_n$ $\quad E[W_i] = 0$, $\quad \text{var}(W_i) = 4$

$$P(\theta - 0.25 \le \bar{x} \le \theta + 0.25) \ge 0.90$$
$$P(-0.25 \le \bar{x} - \theta \le 0.25) \ge 0.90$$
$$P(-\bar{x} - 0.25 \le -\theta \le -\bar{x} + 0.25) \ge 0.90$$
$$P(\bar{x} + 0.25 \ge \theta \ge \bar{x} - 0.25) \ge 0.90$$
$$P(\bar{x} - 0.25 \le \theta \le \bar{x} + 0.25) \ge 0.90$$

(θ: population mean; $\bar{x}$: sample mean; 90% confidence interval for θ)

$(1-\alpha)$ confidence interval for population mean → $\left[\bar{x} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right]$

Here $1 - \alpha = 90\% = 0.9$, $\frac{\alpha}{2} = 0.05$

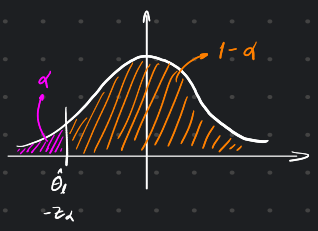
($z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}$: error) $\quad \bar{x} - 0.25 \quad \bar{x} + 0.25$

$$\Rightarrow z_{0.05}\frac{\sigma}{\sqrt{n}} \le 0.25$$
$$1.65 \times \frac{2}{\sqrt{n}} \le 0.25$$
$$n \ge \left(\frac{1.65 \times 2}{0.25}\right)^2$$
$$n \ge 174.24$$
$$n_{min} = 175$$

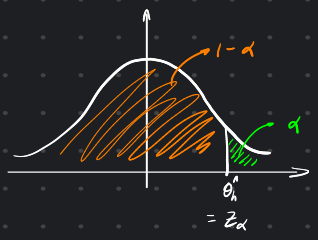
One-sided confidence intervals

Sometimes we need to determine the value so that we can assert, with certain confidence, that μ is atleast as large or as small as this value.

Instead of $P\{\hat{\theta}_1 \le \theta \le \hat{\theta}_2\} \ge 1-\alpha$,
we have $P\{\hat{\theta}_1 \le \theta\} \ge 1-\alpha$, or $P\{\theta \le \hat{\theta}_2\} \ge 1-\alpha$



To find $\hat{\theta}_1$,
area (///) = α



∴ the other one becomes $-z_\alpha$

Let x1, ..., xn be a random sample from a distribution with a known variance var(xi)=σ^2, and unknown mean E[xi]=θ. Find a (1-α) confidence interval (one-sided) for θ, assuming that n is large.

→ $\hat{\theta} = \bar{x} = \frac{x_1 + \dots + x_n}{n}$ $\quad \bar{x}$ is approx. normal (CLT).

$$\Rightarrow \frac{\bar{x}-\theta}{\sigma/\sqrt{n}} \text{ is std. Normal } \sim Z$$

$$\Rightarrow \frac{\sqrt{n}(\bar{x}-\theta)}{\sigma} \sim Z$$

$$P\left\{\frac{\sqrt{n}(\bar{x}-\theta)}{\sigma} < z_\alpha\right\} = 1-\alpha$$

$$\Rightarrow P\left\{\bar{x}-\theta < \frac{\sigma z_\alpha}{\sqrt{n}}\right\} = 1-\alpha$$

$$\Rightarrow P\left\{\theta > \bar{x} - \frac{\sigma z_\alpha}{\sqrt{n}}\right\} = 1-\alpha$$

$$\theta \in \underbrace{\left[\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}\right.}_{\hat{\theta}_1}, \infty)$$

Ny, the other one-sided confidence interval becomes

$$\theta \in \underbrace{(-\infty, \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}]}_{\hat{\theta}_2}$$

Q. When a signal having value μ is transmitted from a locan A. The value received at B is N(μ, 4). To reduce error, the same value is sent 9 times. The successive values received are [5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5].

Find the upper and lower 95% confidence interval estimates for μ.

→ upper $\left[\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}, \infty\right)$ $\quad \alpha = 0.05$, $z_\alpha = 1.65$

$$\equiv \left[9 - (1.65)\frac{(2)}{\sqrt{9}}, \infty\right)$$

$$\equiv [7.9, \infty)$$

lower conf. interval $\equiv (-\infty, 10.1]$

Estimating mean when the variance is unknown

We know that $P\left\{\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \theta \leq \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$

If $P(-1 \leq \theta \leq 1) = 0.9$
then $P(-1.5 \leq \theta \leq 1.5) \geq 0.9$
so if we do it for α, if we expand limits,
$P(m \leq \theta \leq u) \geq 1-\alpha$
↳ so we find an upper bound on σ.
$\sigma \leq \sigma_{max}$.

If σ is unknown, there are 2 approaches:

1. We find an upper bound on σ.

$\sigma \leq \sigma_{max}$

The interval gets bigger: $\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma_{max}}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma_{max}}{\sqrt{n}}\right]$

2. Estimate the values of σ^2.
   Find a relatively good estimated value for σ^2 (say using CLT) and use it.

Q.

(Public Opinion Polling) We would like to estimate the portion of people who plan to vote for Candidate A in an upcoming election. It is assumed that the number of voters is large, and θ is the portion of voters who plan to vote for Candidate A. We define the random variable X as follows. A voter is chosen uniformly at random among all voters and we ask her/him: "Do you plan to vote for Candidate A?" If she/he says "yes," then $X = 1$, otherwise $X = 0$. Then,

$$X \sim \text{Bernoulli}(\theta).$$

Let $X_1, X_2, X_3, \ldots, X_n$ be a random sample from this distribution, which means that the X_i's are i.i.d. and $X_i \sim \text{Bernoulli}(\theta)$. In other words, we randomly select n voters (with replacement) and we ask each of them if they plan to vote for Candidate A. Find a $(1-\alpha)100\%$ confidence interval for θ based on $X_1, X_2, X_3, \ldots, X_n$.

Method 1: upper bound on σ

→ θ : portion of voters for A.

$X \sim \text{Bernoulli}(\theta)$

$X_1, \ldots, X_n$: random sample.

$E[X_i] = \theta$, we want to estimate θ.

Point estimate [MLE] for mean = sample mean of Bernoulli r.v.

The $(1-\alpha)$ confidence interval is $\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$. (assuming n is large)

σ is unknown.

For a Bernoulli r.v.,

$\text{var}(X) = \theta(1-\theta) = \theta - \theta^2$

max value,

$\text{Var}_{max}(x) = 1/4$

$\therefore \sigma_{max} = \frac{1}{2}$

$\frac{d}{d\theta}(\theta - \theta^2) = 0$

$1 - 2\theta = 0$

$\theta = 1/2$

$\text{var}(1/2) = 1/2\,(1-1/2)$

$\text{Var}_{max} = 1/4$

∴ The $1-\alpha$ confidence interval for θ becomes

$$\left[\bar{x} - \frac{z_{\frac{\alpha}{2}}}{2\sqrt{n}}, \bar{x} + \frac{z_{\frac{\alpha}{2}}}{2\sqrt{n}}\right]$$

Q. There are two candidates in a presidential election: Candidate A and Candidate B. Let θ be the portion of people who plan to vote for Candidate A. Our goal is to find a confidence interval for θ. Specifically, we choose a random sample (with replacement) of n voters and ask them if they plan to vote for Candidate A. Our goal is to estimate the θ such that the margin of error is 3 percentage points. Assume a 95% confidence level. That is, we would like to choose n such that

$$P\left(\overline{X}-0.03 \le \theta \le \overline{X}+0.03\right) \ge 0.95,$$

where $\overline{X}$ is the portion of people in our random sample that say they plan to vote for Candidate A. How large does n need to be?

→ margin of error = 3%, $1-\alpha = 0.95$, $\alpha = 0.05$

$$P\left\{\bar{x}-0.03 \le \theta \le \bar{x}+0.03\right\} \ge 0.95$$

$z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \le 0.03$ ← max error

$$\frac{z_{0.025}}{2\sqrt{n}} \le 0.03$$

$$n \ge \left[\frac{1.96}{2(0.03)}\right]^2$$

$$n \ge 1067.11$$

$n_{min} = 1068$

Method 2: Estimate σ

For $x_i \sim$ Bernoulli(θ), $\sigma = \sqrt{\theta(1-\theta)}$

This way just gives us an approx. interval.

$\sigma_{est} = \sqrt{\bar{x}(1-\bar{x})}$

confidence interval: $\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sqrt{\bar{x}(1-\bar{x})}}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sqrt{\bar{x}(1-\bar{x})}}{\sqrt{n}}\right]$

In any general case, we can use the sample variance as the expected value of popl^n variance.

$$s^2 = \frac{1}{n-1} \cdot \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{1}{n-1}\left[\sum x_i^2 - n\bar{x}^2\right]$$

confidence interval: $\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right]$

Q. We collected random sample $x_1, \ldots, x_{100}$. $\bar{x} = 15.6$, $s^2 = 8.4$. Construct an approx. 99% conf. interval for $\theta = E[x_i]$.

→ $\frac{\alpha}{2} = \frac{0.01}{2} = 0.005$ $z_{\frac{\alpha}{2}} = z_{0.005}$

$$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right] = \left[15.6 - 0.75\frac{\sqrt{8.4}}{10}, 15.6 + 0.75\frac{\sqrt{8.4}}{10}\right]$$

$$= [14.85, 16.35]$$

... won't use this unless reqd.

Confidence intervals for normal samples

Theorem:

Let $X_1, \dots, X_n$ be a sample from a normal population with mean μ.
If $\bar{x}$ denotes the sample mean, S^2 the sample variance,
then

$$\frac{\sqrt{n}\,(\bar{x}-\mu)}{S} \sim t_{n-1}$$

t-distribⁿ with $n-1$ degrees of freedom.

Proof:

We know that r.v. $Y = \frac{(n-1)S^2}{\sigma^2}$ is chi-squared with $n-1$ degrees of freedom.

Also, $T_n = \frac{Z}{\sqrt{\chi_n^2/n}} \implies T_{n-1} = \frac{Z}{\sqrt{\chi_{n-1}^2/n-1}}$ (T-dist. in terms of χ^2)

and $\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ is std normal $\sim Z$

$$T = \frac{Z}{\sqrt{Y/n-1}} = \frac{\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}}} = \frac{\sqrt{n}\,(\bar{x}-\mu)}{S}$$

$$= \frac{Z}{\sqrt{\chi_{n-1}^2/n-1}} = T_{n-1}$$

$$\therefore \frac{\sqrt{n}\,(\bar{x}-\mu)}{S} \sim T_{n-1}$$

□

Confidence intervals for normal samples

If the population is normal, we can say that the samples are normal, their sum is normal. So we'll get a more exact interval.

We'll use this:

For a sample taken from a normal distribution (μ, σ^2),
the sample mean $\bar{X} = \frac{X_1 + \ldots + X_n}{n}$ and the sample variance $S^2 = \frac{\sum (X_i - \bar{X})^2}{(n-1)}$
follow the following distributions:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \qquad (n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1} \qquad \frac{\bar{X}-\mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$$

if you don't know pop var, approx to t dist

$$\Downarrow$$

$$\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim Z$$

if you know pop var, approx to std normal

used when we want to approximate variance instead of mean

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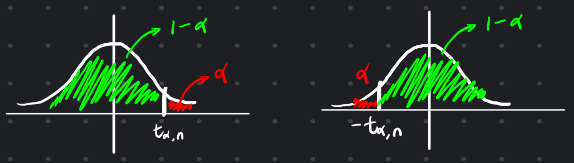
t-score

For α, $0 < \alpha < 1$, let $t_{\alpha,n}$ be defined as:

$$P\{T_n > t_{\alpha,n}\} = \alpha$$

.. similar to $P\{Z > z_\alpha\} = \alpha$

The t-distⁿ is symmetric about the y-axis as well.

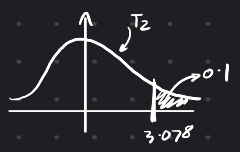


$$P\{T_n > -t_{\alpha,n}\} = 1-\alpha$$

$$-t_{\alpha,n} = t_{1-\alpha,n}$$

(same as z_α)

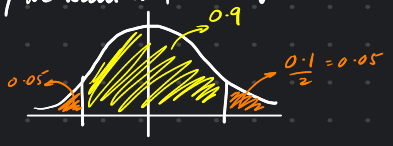
The t-table does not contain cumulative frequencies like the z-table.
It'll give $t_{\alpha,n}$ values directly.



$t_{0.1,1} = 3.078 \Rightarrow P\{T_1 > 3.078\} = 0.1$

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

so say we want a 90% confidence interval.



We'll use the fact that

$$\frac{\bar{X}-\mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$$

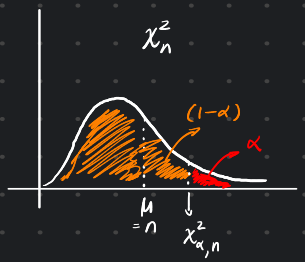
↳ not approx, exact (CLT has not been used)

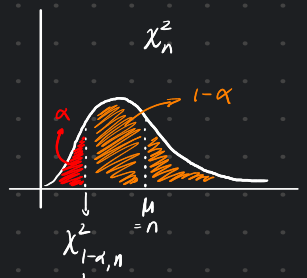
Chi-square score

The Chi-square distribution is not symmetric.

If $X \sim \chi^2_n$, then for any $\alpha \in (0,1)$,

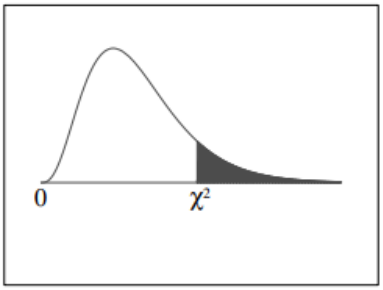
$$P\{\chi^2_n > \chi^2_{\alpha,n}\} = \alpha$$





$$\chi^2_{1-\alpha,n} \neq 1 - \chi^2_{\alpha,n}$$

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Confidence interval for the mean of the normal distribution

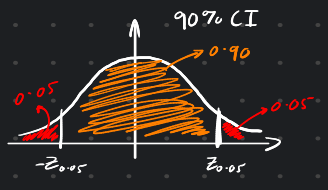
Case: population variance is known.

μ : population mean

Point estimator : sample mean $= \bar{x}$

$$\bar{X} \sim N\left(\mu, \sigma^2/n\right)$$

$$\Rightarrow \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim Z = N(0,1)$$



$$P\left\{-z_{\frac{\alpha}{2}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < z_{\frac{\alpha}{2}}\right\} = 1-\alpha$$

$$P\left\{\bar{X} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$$

$(1-\alpha)\cdot 100\%$ confidence interval for μ is $\left[\bar{X} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right]$.

one-sided confidence intervals :

$$P\left\{\mu > \bar{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$$

$$P\left\{\mu < \bar{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$$

$(1-\alpha)\cdot 100\%$ upper C.I. $= \left[\bar{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}},\ \infty\right)$

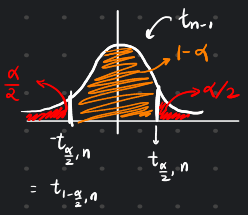
$(1-\alpha)\ 100\%$ lower C.I. $= \left(-\infty,\ \bar{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}\right]$

Case: population variance is NOT known.

we can't use $\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim Z$ as σ is unknown. But we can use the sample variance.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t_{n-1}$$



$$P\left\{-t_{\frac{\alpha}{2},n-1} < \frac{\bar{x}-\mu}{s/\sqrt{n}} < t_{\frac{\alpha}{2},n-1}\right\} = (1-\alpha)$$

$$P\left\{\bar{x}-t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}} < \mu < \bar{x}+t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}\right\} = (1-\alpha)$$

$(1-\alpha)\cdot 100\%$ confidence interval for $\mu = \left[\bar{x}-t_{\frac{\alpha}{2}n-1}\frac{s}{\sqrt{n}},\ \bar{x}+t_{\frac{\alpha}{2}n-1}\frac{s}{\sqrt{n}}\right]$

Q. when a signal having value μ is transmitted from a locan A. The value received at B is $N(\mu,$ unknown variance$)$. To reduce error, the same value is sent 9 times. The successive values received are $[5,\ 8.5,\ 12,\ 15,\ 7,\ 9,\ 7.5,\ 6.5,\ 10.5]$.

Find the 95% confidence interval for μ.

$$\bar{X} = \frac{5+8.5+\cdots+10.5}{9} = 9$$

Instead we could use $\sum(x_i-\bar{x})^2 = \sum x_i^2 - n\bar{x}^2$ as well.

$$s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2 = \frac{1}{8}\left[4^2+0.5^2+3^2+6^2+2^2+0^2+1.5^2+2.5^2+1.5^2\right]$$

$$s^2 = 9.5$$

$$s = 3.082$$

$1-\alpha = 0.95$

$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$

95% confidence interval $\mu \in \left[\bar{x}-t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}},\ \bar{x}+t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}\right]$

$$= \left[9-t_{0.025,8}\frac{3.082}{3},\ 9+t_{0.025,8}\frac{3.082}{3}\right]$$

$$= \left[9-2.306\times\frac{3.082}{3},\ 9+2.306\times\frac{3.082}{3}\right]$$

$$= [6.63,\ 11.37]$$

one-sided confidence intervals for μ:

upper C.I. $= \left[\bar{x}-t_{\alpha,n-1}\frac{s}{\sqrt{n}},\ \infty\right)$

lower C.I. $= \left(-\infty,\ \bar{x}+t_{\alpha,n-1}\frac{s}{\sqrt{n}}\right]$

Prediction intervals

In statistical inference, specifically predictive inference, a **prediction interval** is an estimate of an interval in which a future observation will fall, with a certain probability, given what has already been observed. Prediction intervals are often used in regression analysis.

Suppose $X_1, X_2, \dots, X_n, X_{n+1}$ is a sample from a normal distribution with unknown mean and variance. $X_1, \dots, X_n$ are observed. X_{n+1} is to be predicted.

Point estimator for $X_{n+1} = \dfrac{X_1 + X_2 + \cdots + X_n}{n}$

Now we want to find $(1-\alpha)\cdot 100\%$ C.I. for X_{n+1}

$$\bar{X_n} = \frac{X_1 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Also $X_{n+1} \sim N(\mu, \sigma^2)$

$$E[X_{n+1} - \bar{X_n}] = E[X_{n+1}] - E[\bar{X_n}] = \mu - \mu = 0 \qquad \text{as } E[-\bar{X}] = -E[\bar{X}]$$

$$\begin{aligned} Var(X_{n+1} - \bar{X_n}) &= Var(X_{n+1}) + Var(-\bar{X_n}) \\ &= Var(X_{n+1}) + Var(\bar{X_n}) \\ &= \sigma^2 + \frac{\sigma^2}{n} \end{aligned} \qquad \text{as } Var(-\bar{X}) = (-1)^2 Var(\bar{X}) = Var(\bar{X})$$

$$\therefore \quad X_{n+1} - \bar{X_n} \sim N\left(0, \sigma^2 + \frac{\sigma^2}{n}\right)$$

$$\frac{X_{n+1} - \bar{X_n} - 0}{\sqrt{\sigma^2 + \frac{\sigma^2}{n}}} \sim N(0,1)$$

$$\frac{X_{n+1} - \bar{X_n}}{\sigma\sqrt{1 + \frac{1}{n}}} \sim Z \text{ or } N(0,1)$$

Here σ is not known.

$$\frac{X_{n+1} - \bar{X_n}}{S_n\sqrt{1 + \frac{1}{n}}} \sim t_{n-1}$$

$$\therefore \quad P\left\{-t_{\frac{\alpha}{2}, n-1} < \frac{X_{n+1} - \bar{X_n}}{S_n\sqrt{1 + \frac{1}{n}}} < t_{\frac{\alpha}{2}, n-1}\right\} = 1 - \alpha$$

$$(1-\alpha)\cdot 100\% \text{ C.I. for } X_{n+1} : \left[\bar{X_n} - t_{\frac{\alpha}{2}, n-1}\, S_n\sqrt{1 + \frac{1}{n}},\ \bar{X_n} + t_{\frac{\alpha}{2}, n-1}\, S_n\sqrt{1 + \frac{1}{n}}\right]$$

Q. We observe 7 values from a normal distribution and the sample mean was found to be 6716.57 and the sample variance was found to be 733.97.
Find a 95% confidence interval for the next sample value X8? $\left(t_{0.025, 6} = 2.447\right)$

→

$$X_8 \in \left[\bar{X_7} - t_{\frac{0.05}{2}, 7-1}\, S_7\sqrt{1 + \frac{1}{7}},\ \bar{x} + t_{\frac{0.05}{2}, 7-1}\, S_7\sqrt{1 + \frac{1}{7}}\right]$$

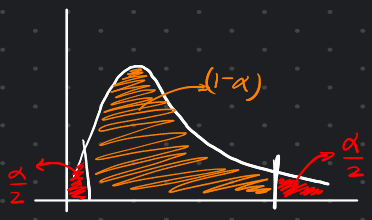
$$= \left[(6716.57) - (2.447)(733.97)\sqrt{\frac{8}{7}},\ (6716.57) + (2.447)(733.97)\sqrt{\frac{8}{7}}\right] = [4796, 8637]$$

$X_8 \in [4796, 8637]$ with 95% confidence.

Confidence interval for the variance of a normal distribution

If $X_1, \ldots, X_n$ is a sample from a normal distribution having unknown parameters (μ, σ^2), we can construct $(1-\alpha)\cdot 100\%$ C.I. for σ^2 using the fact:

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$



$$P\left\{ \chi^2_{1-\frac{\alpha}{2},n-1} < (n-1)\frac{S^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2},n-1} \right\} = (1-\alpha)$$

$$\Rightarrow P\left\{ \frac{1}{\chi^2_{1-\frac{\alpha}{2},n-1}} > \frac{\sigma^2}{S^2(n-1)} > \frac{1}{\chi^2_{\frac{\alpha}{2},n-1}} \right\} = (1-\alpha)$$

$$\Rightarrow P\left\{ \frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}} \right\} = (1-\alpha)$$

$(1-\alpha)\cdot 100\%$ interval for σ^2: $\sigma^2 \in \left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}} \right]$

$(1-\alpha)\cdot 100\%$ one-sided C.I.

upper: $\sigma^2 \in \left[\frac{(n-1)S^2}{\chi^2_{\alpha,n-1}}, \infty \right)$

lower: $\sigma^2 \in \left(-\infty, \frac{(n-1)S^2}{\chi^2_{1-\alpha,n-1}} \right]$

Q. A farmer weights 10 randomly chosen watermelons from his farms and obtains the following values (in lbs):
7.72, 9.58, 12.38, 7.77, 11.27, 8.8, 11.1, 7.8, 10.17, 6.0
Assuming that the weight of a watermelon is normally distributed (μ, σ^2),
a. Find a 95% C.I. for μ.
b. Find a 95% C.I. for σ^2.

$\rightarrow \bar{x} = \frac{x_1 + \dots + x_n}{n} = 9.26 \qquad s^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2 = 3.96$

a) $(1-\alpha) \cdot 100\%$ C.I. for μ:

$(1-\alpha) = 0.95$

$\frac{\alpha}{2} = 0.025$

$$\mu \in \left[\bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} \right]$$

$$\equiv \left[9.26 - t_{0.025, 10-1} \frac{\sqrt{3.96}}{\sqrt{10}}, 9.26 + t_{0.025, 10-1} \frac{\sqrt{3.96}}{\sqrt{10}} \right]$$

$\mu \in [7.84, 10.68]$ with 95% confidence using $t_{0.025, 9} = 2.262$

b) $(1-\alpha) \cdot 100\%$ C.I. for σ^2:

$$\sigma^2 \in \left[\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right]$$

$$\equiv \left[\frac{9(3.96)}{(19.02)}, \frac{9(3.96)}{(2.70)} \right]$$

$\chi^2_{0.025, 9} = 19.02$

$\chi^2_{0.975, 9} = 2.70$

$\sigma^2 \in [1.87, 13.2]$ with 95% confidence.

If $X_1, X_2, ..., X_n \sim \mathcal{N}(\mu, \sigma^2)$, $\bar{x} = \frac{\Sigma_{i=1}^n}{n}$, $s^2 = \frac{\Sigma_{i=1}^n (x_i - \bar{x})^2}{n-1}$, then

Parameter	Assumption	$(1-\alpha) \times 100\%$ C.I.	Lower Interval	Upper Interval
μ	σ^2 is known	$[\bar{x} - z_{\alpha/2}(\frac{\sigma}{\sqrt{n}}), \bar{x} + z_{\alpha/2}(\frac{\sigma}{\sqrt{n}})]$	$(-\infty, \bar{x} + z_{\alpha}(\frac{\sigma}{\sqrt{n}})]$	$[\bar{x} - z_{\alpha}(\frac{\sigma}{\sqrt{n}}), \infty)$
μ	σ^2 is unknown	$[\bar{x} - t_{\alpha/2,n-1}(\frac{s}{\sqrt{n}}), \bar{x} + t_{\alpha/2,n-1}(\frac{s}{\sqrt{n}})]$	$(-\infty, \bar{x} + t_{\alpha,n-1}(\frac{s}{\sqrt{n}})]$	$[\bar{x} - t_{\alpha,n-1}(\frac{s}{\sqrt{n}}), \infty)$
σ^2	μ is unknown	$[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}]$	$(-\infty, \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}]$	$[\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}}, \infty)$